

ASSIGNMENT: THERMODYNAMICS

ANSWER ALL THE QUESTIONS

DATE: 22/2/26

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1. A mass of **0.20 kg** of an ideal gas with constant specific heats, having a volume of **0.15 m³** and a temperature of **10 °C**, is adiabatically compressed to a volume of **0.046 m³** and a pressure of **4 bar** in a piston-cylinder arrangement. Find (i) gas constant, (ii) molecular mass of the gas, (iii) ratio of the specific heats, i.e., c_p/c_v (iv) two specific heats (v) change in internal energy (vi) find the work of compression. Assume the compression process to be quasistatic and frictionless.
2. An imaginary engine receives heat and does work on a slowly moving piston at such a rate that the cycle of operation at **1 kg** of working fluid can be represented as a circle **25 cm** in diameter on P-v diagram on which **1cm parallel to y-axis=500 kPa** and **1 cm on x axis = 0.2 m³/kg**. How much work is done by each kg of working fluid for each cycle of operation and heat input in cycle. If the heat rejected by the engine in a cycle is **30000kJ/kg** of working fluid, what would be its thermal efficiency.
3. A fluid is confined in a cylinder by a spring-loaded, frictionless piston so that the pressure in the fluid is a linear function of the volume, as given by ($p = a + bV$). The internal energy of the fluid is given by the following equation $U = 34 + 3.15pV$ where U is in kJ, p is in kPa and V in cubic meter. If the fluid changes from an initial state of **170 kPa, 0.03 m³** to final state of **400 kPa, 0.06 m³**. With no work other than that done on the piston, find the direction and magnitude of the work and heat interactions.
4. Find out the following table for a substance water:

	P(kPa)	T°C	v(m ³ /kg)	x
a	500			
b	500		0.2	
c	1400	200		
d		300		0.8

(Use steam Table from any book)

Represent these points on P-v and T-v Plots